

STUDENT MANAGEMENT SYSTEM

A Project Report

Submitted in partial fulfillment of the
Requirements for the award of the Degree of
BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

By

Mrs. Satpute Ruchi Gajanan

Seat number: 3024867

Under the Esteemed Guidance of

Mr. Amit S. Shevade

M.Sc. (CS), MBA (IT)



DEPARTMENT OF INFORMATION TECHNOLOGY

N.S. SOCIETY'S

D.B.J. COLLEGE

NAAC CYCLE RECREDITED GRADE 'A'CGPA: 3.14)

(Affiliated to University of Mumbai)

CH IPLUN, 415605

MAHARASHTRA

SEMESTER-V

2024-2025

D.B.J. COLLEGE
(Affiliated to University of Mumbai)
CH IPLUN- MAHARASHTRA-415605
DEPARTMENT OF INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the project entitled, “**STUDENT MANAGEMENT SYSTEM**”, is bonafied work of **SATPUTE RUCHI GAJANAN** bearing Seat No: _____ submitted in partial fulfillment of the requirements for the award of degree of BACHLOER OF SCIENCE in INFORMATION TECHNOLOGY from University of Mumbai.

Internal Guide

Coordinator

External Examiner

Date:

College Seal

ORGANIZATION CERTIFICATE

Yash English Academy

Dhamanvane, Chiplun - 415605

District: Ratnagiri

This is to certify that Miss. SATPUTE RUCHI GAJANAN student of D.B.J. College,
Chiplun, has successfully completed her project work entitled

“STUDENT MANAGEMENT SYSTEM”

As a part of her curriculum for the award of degree of BACHLOER OF SCIENCE in
INFORMATION TECHNOLOGY from

University of Mumbai.

Her work was satisfactory during the project period.

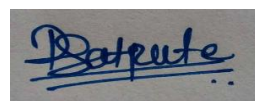
Authorized Signature

DECLARATION

I hereby declare that the project entitled, “**Student Management System**” done at **D.B.J. College, Chiplun**, has been in any case duplicated to submit to any other university for the award of any degree.

To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfillment of the requirements for the award of degree of **BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)** to be submitted as final semester project as part of our curriculum.

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Satpute' with a stylized flourish underneath.

Mrs. Ruchi Gajanan Satpute.
TY. BSc (Information Technology)

ACKNOWLEDGEMENT

It's my great pleasure to take this opportunity and sincerely thanks to all of those, who shown me the way to successful project and help me a lot during the completion of my project.

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ABSTRACT

Most of the Academic institutions face difficulty in managing records of students, attendance, accounts, etc., and track the information of their interest as they still rely on paperwork and manual processes.

A Student Management System is an environment that manages all the data of the students who are studying in an educational institution. This data is computerized through an automated system. Here, computerization is more advantageous than the usual method.

Thus, a student management system offers many benefits to an educational institution. It allows teachers to easily change and access student data, and parents can easily focus on children with a clear environment to meet state level compliance and other regulatory requirements.

This system we have created also facilitates the entry, maintenance and viewing and of all authorized student details. Here, we mainly focus on the examination of students, their subjects, the registration which conducts the registration process, the examination department which conducts the examinations.

The pandemic has changed the way the world used to operate. The biggest change has possibly been experienced by educational institutions. This has made it mandatory for all schools to use intelligently designed school management software.

A well-designed school management system will reduce the workload of staff, help both students and teachers to save time, enable cost reduction, and also enhance data security. All these will eventually increase your institution's cost-effectiveness and productivity.

TABLE OF CONTENTS

Chapter 1: Introduction	9
1.1 Background	11
1.2 Problem Statement	12
1.3 Objective	13
1.4 Scope	14
1.5 Organization Report	15
 Chapter 2: Survey of Technologies	 16
2.1 Software Description	16
2.2 Why using PHP?	17
2.3 Why using MYSQL?	17
 Chapter 3: System Analysis	 18
3.1 Existing System	18
3.2 Proposed System	19
3.3 Feasibility Study	20
3.4 Requirement Analysis	22
3.5 Software and Hardware Requirements	24
3.6 Fact Finding Techniques	25
 Chapter 4: System Design	 27
4.1 Basic Module	27
4.2 Input Design	29
4.3 Output Design	30

4.4 Data Design.....	31
4.5 Gantt Chart.....	35
4.6 UML Diagrams	37
Chapter 5: Implementation and Testing	43
5.1 Implementation Approach	43
5.2 Testing Approach.....	44
5.2 Validation Check	46
5.2 Test Cases	47
Chapter 6: Screen Layouts.....	50
Chapter 7: Source Code	57
Chapter 8: Conclusions	64
8.1 Conclusion	64
8.2 Future Work.....	65
Chapter 9: References	66

List of Figures

Fig. 4.6.1 Class Diagram.....	37
Fig. 4.6.2 E-R Diagram	38
Fig. 4.6.3 Use Case Diagram	39
Fig. 4.6.4 Data Flow Diagram	41
Fig. 4.6.5 Sequence Diagram.....	42

List of Tables

Table 4.4.1 Login Record Table.....	32
Table 4.4.2 Student Details Table.....	32
Table 4.4.3 Attendance Record Table	33
Table 4.4.4 Student Marks Table	33
Table 4.4.5 Timetable Details Table.....	34

Chapter 1: Introduction

1.1 Background

Storing the details of students in school or institute has always been a time-taking and hectic task. It becomes very difficult for an individual to mark the attendance and details of a large number of students. Students often “bunk” classes for various reasons. It becomes difficult to track these absences when a friend is fitted with the responsibility of standing in for them as a proxy during the rollcall.

Moreover, many teachers often skip the roll call too, either due to a shortage of teaching time or laziness. This not only affects the academic performance of the student but can also cause the morale of the professor to be affected. The parents at home don't even know if their child is going to school/college or not.

There's no way for them to know until the school/college directly contacts them and tells them about it in details. The management system has to be changed in order to make the task easy, to make the students attend their lectures, and to give parents a feeling of relief by letting them know that their child is focusing on their studies. The best way of marking the details of the students and making it more accessible is by using a website.

This website is very easily available online and give us a number of benefits due to which they are recommended more. Mostly they are used by large coaching institutes for keeping a note of the number of classes attended by students, their timings, and other details.

Almost all Schools and Colleges still use this method to track student details. Yet most modern Schools and Colleges don't use it anymore because it's inconvenient and requires to have a responsible person for filling it in.

Attendance and exams of a large number of students can be managed at the same time and that too effectively. Hence student management becomes quite easier. This is very easy to handle other tasks also and therefore no specific requirements are needed for an individual to use them.

1.2 Problem Statement

The problem occurred before having computerized system includes:

- File lost when computerized system is not implemented, file is always lost because of human behaviour, due to some human error there may be a loss of records.
- File gets damaged when a computerized system is not there, some cases like due to natural disasters, fire, floods etc.
- Difficulty to search record when there is no computerized system there is always difficulty in searching of records if the records are large in number.
- Parents and students often had limited access to their children's information, and even staff members had difficulty accessing data quickly.
- Data was often fragmented across different departments or individuals, making it difficult to get a comprehensive view of a student's progress or situation.
- Space consuming, after the number of records become large the space for physical storage of file and records also increases.
- Without the computerized system it becomes very cost consuming as there is no computerized system to add each record, paper will be needed this will increase the cost management.

1.3 Objectives

The goal of project is basically very simple but also important and wants to offer a particularly simple solution to the masses in a particularly important way, or so they for all intents and purposes thought. Definitely provide them with an ethical system to generally make their leisure time generally more fluid and significantly kind of more important in a subtle way.

Users really can connect to the system in different way, which will help them enormously. The main objective of the Student Management System is to manage the details of Profiles, Courses, Logins, Exams. It manages all the information about each student. The control of this system is given to teacher. The project is totally built at administrative end and thus only the administrator is guaranteed access.

Functionalities provided by Student Management System:

- Simplifying all Tasks.
- Better communication.
- Easy access to all.
- Easy to manage the data in efficient manner.
- Complete tracking of all the students with their proper details.
- Timetable can be managed by the teachers very easily.

These are the functionalities making the application very beneficial for all of the schools, colleges etc. This application has some others useful cases also. Like by using this we are saving huge paperwork. Anything we do to save paper will help reduce the amount of trash going into landfills and it will also reduce energy use and pollution associated with manufacturing, transporting and recycling new paper products.

1.4 Scope

The system is aimed at total user-friendly as well as efficient management of varied tasks. These tasks may range from registering new students, managing timetable, examination management to all the essential features necessary for making the administrative division of school effective.

In modern times, facilities offered by schools are not limited to basic functioning instead; the authorities have been looking for advanced system. In order to cope up with all these factors, the school management system was developed and nowadays, it has even been recognized by most of the Indian schools or colleges. As a matter of fact, this system based on smart technology has become an integral part of many schools.

The school management system is basically manufactured to compile all manual activities of administrative importance in the form of software. This software further makes it easier for officials to finish off their work in a lesser span of time.

Most of all, the mechanism of software is easy to understand that even if any school is utilizing it for the first time, the users will not have to toil hard to learn its function.

On the other hand, there is a vast range of apps that are included in this software for different streams of management in any school. For instance, if you have purchased a school app or similar software, then its various modules will make your works simpler yet very accurate.

1.5 Organisation Report

Name: Yash English Academy, Chiplun

Address: Kapila Park, Dhamanvane, Chiplun, Maharashtra-415605

Parent Institution: Shree Sadguru Education Society

Affiliated to: Central Board of Secondary Education (CBSE)

Taluka: Chiplun

District: Ratnagiri

Chapter 2: Survey of Technologies

2.1 Software Description

Visual studio Code

Visual Studio Code combines the simplicity of a source code editor with powerful developer tooling, like IntelliSense code completion and debugging.

First and foremost, it is an editor that gets out of your way. The delightfully frictionless edit-build-debug cycle means less time fiddling with your environment, and more time executing on your ideas.

Languages

- **HTML**

HTML stands for **Hypertext Markup Language**. It allows the user to create and structure sections, paragraphs, headings, links, and blockquotes for web pages and applications.

HTML is not a programming language, meaning it doesn't have the ability to create dynamic functionality. Instead, it makes it possible to organize and format documents, similarly to Microsoft Word.

- **CSS**

Cascading Style Sheets, fondly referred to as CSS, is a simple design language intended to simplify the process of making web pages presentable.

CSS handles the look and feel part of a web page. Using CSS, you can control the Color of the text, the style of fonts, the spacing between paragraphs, how columns are sized and laid out, what background images or colours are used, layout designs, variations in display for different devices and screen sizes as well as a variety of other effects.

- **JavaScript**

JavaScript is a dynamic computer programming language. It is lightweight and most commonly used as a part of web pages, whose implementations allow client-side script to interact with the user and make dynamic pages. It is an interpreted programming language with object-oriented capabilities.

2.2 Why Using PHP?

There are a lot of reasons to know and love PHP, probably the most potent and valid of which is this: it's used and runs everywhere the web does. Your cheap little \$3 per month hosting account may let you run a web application in Python or Ruby if you shop carefully. But it'll definitely run PHP. This means that you can count on it wherever you are.

And because it runs everywhere, and is easy to get started with, a lot of very popular software is written in PHP.

2.3 Why using MYSQL?

Many of the world's largest and fastest-growing organizations including Facebook, Google, Adobe, Alcatel Lucent and Zappos rely on MySQL to save time and money powering their high-volume Web sites, business-critical systems and packaged software.

Since then, the performance & scalability, reliability, and ease of use of the world's most **popular** open-source database, characteristics that made **MySQL** the #1 choice for web applications, have relentlessly been improved.

Chapter 3: System Analysis

3.1 Existing System

System Analysis is a detailed study of the various operations performed by a system and their relationships within and outside of the system. Here the key question is- what all problems exist in the present system? What must be done to solve the problem? Analysis begins when a user or manager begins a study of the program using existing system.

During analysis, data collected on the various files, decision points and transactions handled by the present system. The commonly used tools in the system are Data Flow Diagram, interviews, etc. Training, experience and common sense are required for collection of relevant information needed to develop the system.

The success of the system depends largely on how clearly the problem is defined, thoroughly investigated and properly carried out through the choice of solution. A good analysis model should provide not only mechanisms of problem understanding but also the framework of the solution. Thus, it should be studied thoroughly by collection data about the system. Then the proposed system should be analysed thoroughly accordance with the needs.

System analysis can be categorized into following parts:

- System planning and initial investigation.
- Information gathering.
- Applying analysis tools for structured analysis.
- Feasibility study.
- Cost/ Benefit analysis.

In the current system we need to keep a number of records to the student and want to enter the details of the student and marks manually. In this system, only the teacher or the school authority views the mark of the student and they want to enter the details of the student. This is time consuming and has much cost.

3.2 Purposed System

In our proposed system we have the provision for adding the details of the students by the admin. So, the details can't be misused. Another advantage of the system is that it is very easy to edit the details of the students when it found unnecessary.

Our proposed system has several advantages:

- User friendly interface.
- Fast access to database.
- More Storage Capacity.
- Less Error.
- Search Facility.
- Look and Feel Environment.

All the manual difficulties in managing the student details in a school or college have been rectified by implanting computerization.

3.3 Feasibility Study

To evaluate feasibility, a feasibility study is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software. Information such as resource availability, cost estimation for software development, benefits of the software to the organization after it is developed and cost to be incurred on its maintenance are considered during the feasibility study. The objective of the feasibility study is to establish the reasons for developing the software that is acceptable to users, adaptable to change and conformable to established standards.

System analysis

Gather, analyse, and validate the information. Define the requirements and prototypes for new system.

Hardware and software study

The hardware and software is developed or maintained for the smooth running of the project.

System design

The design phase comes after a good understanding of customer's requirements, this phase defines the elements of a system, the components, the security level, modules, architecture and the different interfaces and type of data that goes through the system.

System testing and implementation

Testing is becoming more and more important to ensure customer's satisfaction, and it requires no knowledge in coding, hardware configuration or design. In this phase, the system is ready to be deployed and installed in customer's premises.

Evaluation

This phase identifies whether the system meets the initial requirements and objectives. This is when the system is evaluated for weaknesses. The objective of the evaluation phase of the systems development life cycle is to deploy the system and train the system end users.

Maintenance and modification

In this phase, periodic maintenance for the system will be carried out to make sure that the system won't become obsolete, this will include replacing the old hardware and continuously evaluating system's performance, it also includes providing latest updates for certain components to make sure meets the right standards and the latest technologies to face current security threats.

3.4 Requirement Analysis

Gathering requirements before beginning any project, the needs must be gathered and the viability checked. If the requirements are doable, the project can be continued.

Stakeholders gather all of the requirements needed to build and implement the project during this phase, which are then communicated to the project's developer and designer.

The requirements for this project, which will culminate in a web application, are divided into six categories: Student Management Service, Course Management Service, Attendance Management Service, Administration Management Service, Document Management Service, and an Employee Management Service.

1. Student Management Service:

The student can use this service to check their attendance, progress report, and results, as well as send requests for any required documents, view notifications, examine timetables, and view and submit assignments. Students have the opportunity to provide comments on the teacher's performance in class.

2. Administration Management Service:

The administrator will have full access to all resources in this service. The administrator has the ability to add, update, and delete student information. Also, the administrator controls the overall the website's behaviour.

3. Teacher management Service:

Teachers need to be able to resolve conflicts effectively, both within the classroom and with parents or other staff. The teacher can use this service to check student attendance, progress, set notifications, exam timetables.

4. Course Management Service:

The administrator will be able to add, amend, and delete courses using this service. The administrator will also be able to add, alter, and delete the course's subjects. Only the administrator's courses are visible to the teacher, guardian, and students.

5. Attendance Management Service:

Using this service, administrators will be able to submit, edit, and delete student attendance based on the course and class they are enrolled in. The attendance is only visible to the teacher, guardian, and pupils.

6. Timetable Management Service:

A student timetable is a crucial component of student management, providing a structured schedule for classes, exams, and other academic activities. It facilitates effective time management, ensures efficient resource allocation, and promotes a smooth learning environment for students.

7. Exam Management Service:

Exam records are a crucial component of student management systems, allowing educational institutions to track and manage student academic performance. These records typically include exam results, grades, and other relevant information, accessible through student portals or dashboard.

3.5 Hardware Requirements and Software Requirements

Hardware Requirements

- Processor minimum Intel Core i3 or higher recommended.
- At least 2 or 3 GB RAM to run without any risks.
- Support appropriate drivers are installed.
- System Type: 64bit Operating System.
- Display: 1280 x 800 Least Screen resolution.

Software Requirements

1. HTML, CSS, JavaScript
2. PHP
3. Apache Server
4. MySQL Database
5. Any Operating System
6. Any Web Browser
7. MySQL database will be accessed by XAMPP php MyAdmin.

3.6 Fact Finding Techniques

Basically, these techniques help you gather all the important details you need before you design and build the system. Here are the main techniques you can use:

1. Interviews

- Talk directly with principals, teachers, admin staff, and maybe students. Understand how they currently manage student data, their pain points, and what features they really want.
- Questions:
 - i. How do you currently register new students?
 - ii. What problems do you face in managing student attendance?
 - iii. What reports do you often need?

2. Questionnaires / Surveys

- Create a set of structured questions and send them out. Get feedback from a larger group quickly.
- Questions:
 - i. On a scale of 1–5, how easy is the current student database to use?
 - ii. What features would you like to see in the new system?

3. Observation

- Visit the school office/classroom and *watch* how they manage student information in real-time. See the actual workflow, spot issues they might not even mention.
- Examples:
 - i. Observe how a new admission form is processed.
 - ii. See how attendance is marked and recorded.

4. Document Analysis

- Review existing documents and forms (admission forms, report cards, attendance sheets). Understand what information is captured and how it's organized.
- Examples:
 - i. Study the fields in the admission form.
 - ii. Analyse current grade reports.

5. Prototyping

- Create a rough mock-up (even just on paper) of how the new system might look and work. Get early feedback and refine the project before full development.
- Examples:
 - i. Sketch a sample student profile page and ask, "Would this layout work for you?"

Chapter 4: System Design

4.1 Basic Modules

1. Login Module

- Login form: Users enter their username and password to access the system.
- Role-based authentication: Different roles such as students, teachers and administrators are provided with different access privileges.
- User Profile: Once logged in, users can access their profile and personal information.

2. Home Page Module

- Role-specific views: The home page content changes based on whether the user is a student, teacher, or administrator.
- Navigation menu: Links to various modules like grades, attendance, and more.
- Notifications and Alerts: Displays upcoming deadlines, grades updates, or attendance issues for students or faculty.

3. Student Registration Module

- Personal Details: Students can update their personal information.
- Course Enrolment: Students can view available courses and enrol in them.
- Confirmation & Schedule: Displays enrolled courses and class schedule.

4. Attendance Management Module

- Manual Entry: Allows teachers to mark attendance manually in case of absence of automated systems.
- Absence Alerts: Notifies students or faculty if attendance falls below a threshold.
- Attendance Reports: Generates reports for both students and faculty showing attendance trends.

5. Grade Management Module

- Grade Entry: Faculty can enter grades for assignments, exams.
- Grade Calculation: Calculates final grades based on set criteria.
- Grade Reports: Generates individual grade reports for students.

6. Timetable and Scheduling Module

- Timetable Generation: Generates student timetables based on course enrolment.
- Class Scheduling: Assigns specific times and rooms for each class.
- Conflict Resolution: Ensures there are no overlapping courses for students.

7. Document Management Module

- File Uploads: Allows students to upload assignments, projects, and other documents.
- Document Storage: Provides a secure place for storing academic records, certificates, and other official documents.
- Access Control: Restricts access to sensitive documents based on user roles.

4.2 Input Design

Input design is the process of converting user-oriented input to a computer-based format. Input design is a part of overall system design, which requires very careful attention. Often the collection of input data is the most expensive part of the system. The main objectives of the input design are:

1. Produce cost effective method of input.
2. Achieve highest possible level of accuracy.
3. Ensure that the input is acceptable to and understood by the staff.

Input Data:

The goal of designing input data is to make entry easy, logical and free from errors as possible. The entering data entry operators need to know the allocated space for each field; field sequence and which must match with that in the source document. The format in which the data fields are entered should be given in the input form. Here data entry is online; it makes use of processor that accepts commands and data from the operator through a key board. The input required is analysed by the processor. It is then accepted or rejected. Input stages include the following processes:

- Data Recording
- Data Verification
- Data Control
- Data Transmission
- Data Correction.

Input files can exist in document form before being input to the computer. Input design is rather complex since it involves procedures for capturing data as well as inputting it to the computer.

4.3 Output Design

Outputs from computer systems are required primarily to communicate the results of processing to users. They are also used to provide a permanent copy of these result for latter consultation. Computer output is the most important and direct source of information to the users.

Designing computer output should proceed in an organized well throughout the manner. The right output must be available for the people who find the system easy to use. The outputs have been defined during the logical design stage. If not, they should define at the beginning of the output designing terms of types of output connect, format, response etc,

Various types of outputs are:

- External outputs
- Internal outputs
- Operational outputs
- Interactive outputs
- Turn around outputs

All screens are informative and interactive in such a way that the user can full fill his requirements through asking queries.

4.4 Data Design

The data in the system has to be stored and retrieved from database. Designing the database is part of system design. Data elements and data structures to be stored have been identified at analysis stage. They are structured and put together to design the data storage and retrieval system.

A database is a collection of interrelated data stored with minimum redundancy to serve many users quickly and efficiently. The general objective is to make database access easy, quick, inexpensive and flexible for the user. Relationships are established between the data items and unnecessary data items are removed. Normalization is done to get an internal consistency of data and to have minimum redundancy and maximum stability. This ensures minimizing data storage required, minimizing chances of data inconsistencies and optimizing for updates.

Database Tables

A table is a collection of related data held in a table format within a database. It consists of columns and rows.

In relational databases, and flat file databases, a table is a set of data elements (values) using a model of vertical columns (identifiable by name) and horizontal rows.

A table has a specified number of columns, but can have any number of rows. Each row is identified by one or more values appearing in a particular column subset. A specific choice of columns which uniquely identify rows is called the primary key.

1. Login Record Table

Sr. No.	Column Name	Data Type	Size	Constraints
1.	Email	Varchar	100	Not Null
2.	Password	Varchar	100	Not Null

Table 4.4.1 Login Record Table

2. Student Details Table

Sr. No.	Column Name	Data Type	Size	Constraints
1.	Id	Int	40	Primary key
2.	First Name	Varchar	100	Not null
3.	Last Name	Varchar	100	Not Null
4.	Gender	Varchar	10	Not null
5.	Class	Varchar	20	Not Null
6.	Email	Varchar	100	Not Null
7.	Phone Number	varchar	15	Not Null

Table 4.4.2 Student Details Table

3. Attendance Record Table

Sr. No.	Column Name	Data Type	Size	Constraints
1.	Id	Int	40	Not Null
2.	Attendance	Varchar	10	Not null
3.	Class	Varchar	30	Not Null
4.	Section	Varchar	5	Not null
5.	Date	Datetime	-	Not Null

Table 4.4.3 Attendance Record Table

4. Student Marks Table

Sr. No.	Column Name	Data Type	Size	Constraints
1.	Exam id	Varchar	40	Not Null
2.	Subject	Varchar	255	Not null
3.	Student id	Varchar	40	Not Null
4.	marks	Varchar	10	Not null

Table 4.4.4 Student Marks Table

5. Timetable Details Table

Sr. No.	Column Name	Data Type	Size	Constraints
1.	Class	Varchar	45	Not Null
2.	Section	Varchar	10	Not null
3.	Start Time	Varchar	20	Not Null
4.	End Time	Varchar	20	Not null
5.	Timestamp	Datetime	-	Not Null

Table 4.4.5 Timetable Details Table

4.5 Gantt Chart

Sr. No	Phase		Expected Date of Completion	Actual Date of Completion	Remark
1.	Introduction				
	A.	Background	19/07/2024	25/07/2024	
	B.	Problem Statement	22/07/2024	26/07/2024	
	C.	Objectives	22/07/2024	30/07/2024	
	D.	Scope	30/07/2024	04/08/2024	
	E.	Organization Report	30/07/2024	06/08/2024	
2.	Survey Of Technologies				
	A.	Software Description	09/08/2024	16/08/2024	
	B.	PHP	13/08/2024	16/08/2024	
	C.	MySQL	20/08/2024	24/08/2024	
3.	System Analysis				
	A.	Existing system	09/08/2024	10/08/2024	
	B.	Proposed system	10/08/2024	16/08/2024	
	C.	Requirement Analysis	10/08/2024	16/08/2024	
	D.	Hardware and Software Requirements	22/08/2024	24/08/2024	
	E.	Fact Finding Techniques	22/08/2024	24/08/2024	

4.	System Design				
	A.	Basic Modules	25/08/2024	02/09/2024	
	B.	Input Design	03/09/2024	05/09/2024	
	C.	Output Design	03/09/2024	06/09/2024	
	D.	Data Dictionary	06/09/2024	06/09/2024	
	E.	Gantt Chart	06/09/2024	10/09/2024	
	F.	UML Diagrams	06/09/2024	10/09/2024	
5.	Implementation and Testing				
	A.	Implementation Approach	15/09/2024	25/09/2024	
	B.	Testing Approach	26/09/2024	06/10/2024	
	C.	Validation Check	10/10/2024	20/10/2024	
	D.	Test Cases	25/10/2024	15/11/2024	
6.	Screen Layouts		17/11/2024	30/11/2024	
7.	Source Code		09/12/2024	16/01/2025	
8.	Conclusions		19/01/2025	25/01/2025	
9.	References		29/01/2025	08/02/2025	

4.6 UML Diagrams

- **Class Diagram:**

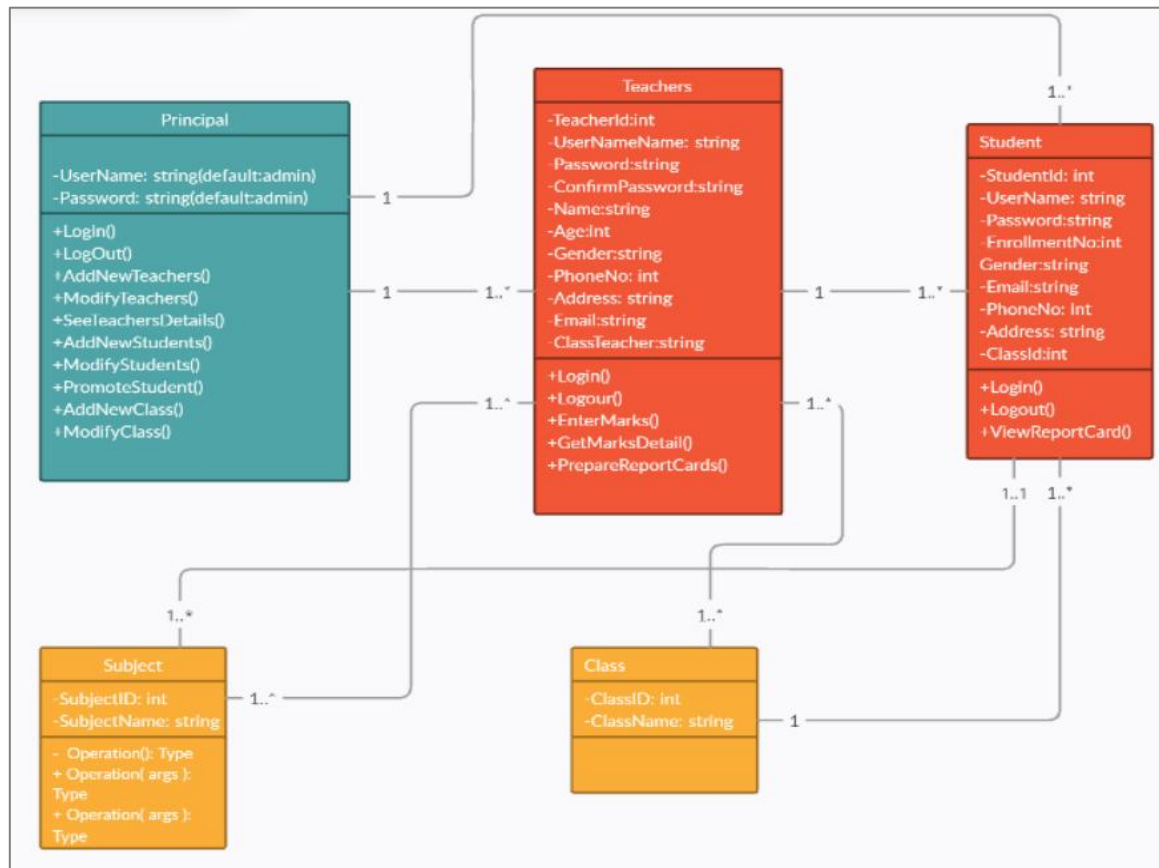


Fig. 4.6.1: Class Diagram for Student Management System

A class diagram is a type of static structure diagram within the Unified Modelling Language (UML) that describes the structure of a system. This diagram outlines the key entities within the system, known as classes, and the relationships between these classes. Each class is represented by a box which contains three compartments:

1. The top compartment contains the name of the class.
2. The middle compartment lists the attributes or properties of the class.
3. The bottom compartment includes the methods or operations the class can execute.

- **E-R Diagram:**

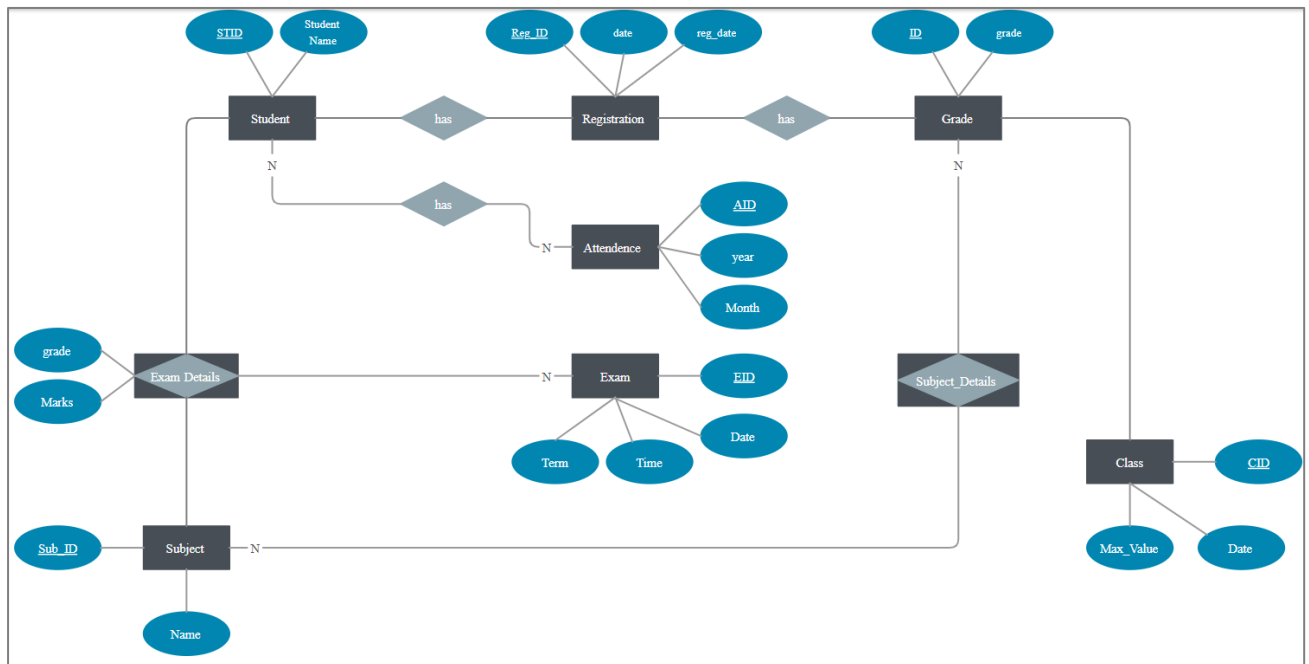


Fig. 4.6.2: E-R Diagram for Student Management System

ER diagram for a student management system shows how the databases are connected to each other. We can also create an ER diagram by drawing the figure of a different part of the student management system, their properties, and how they perform their task.

Before explaining the ER diagram of the student management system, the following labels are used. These are as follows.

1. **Entity:** In a relational database, it represents the table in which all the row represents an entity instance.
2. **Attribute:** In the Database management system, the attributes refer to the database component. The database component includes a table. The attribute also refers to the field of the database.
3. **Relationship:** A relationship is present between the two relational database tables. This is possible when the foreign key of one table references the primary key of another table.

- **Use Case Diagram:**

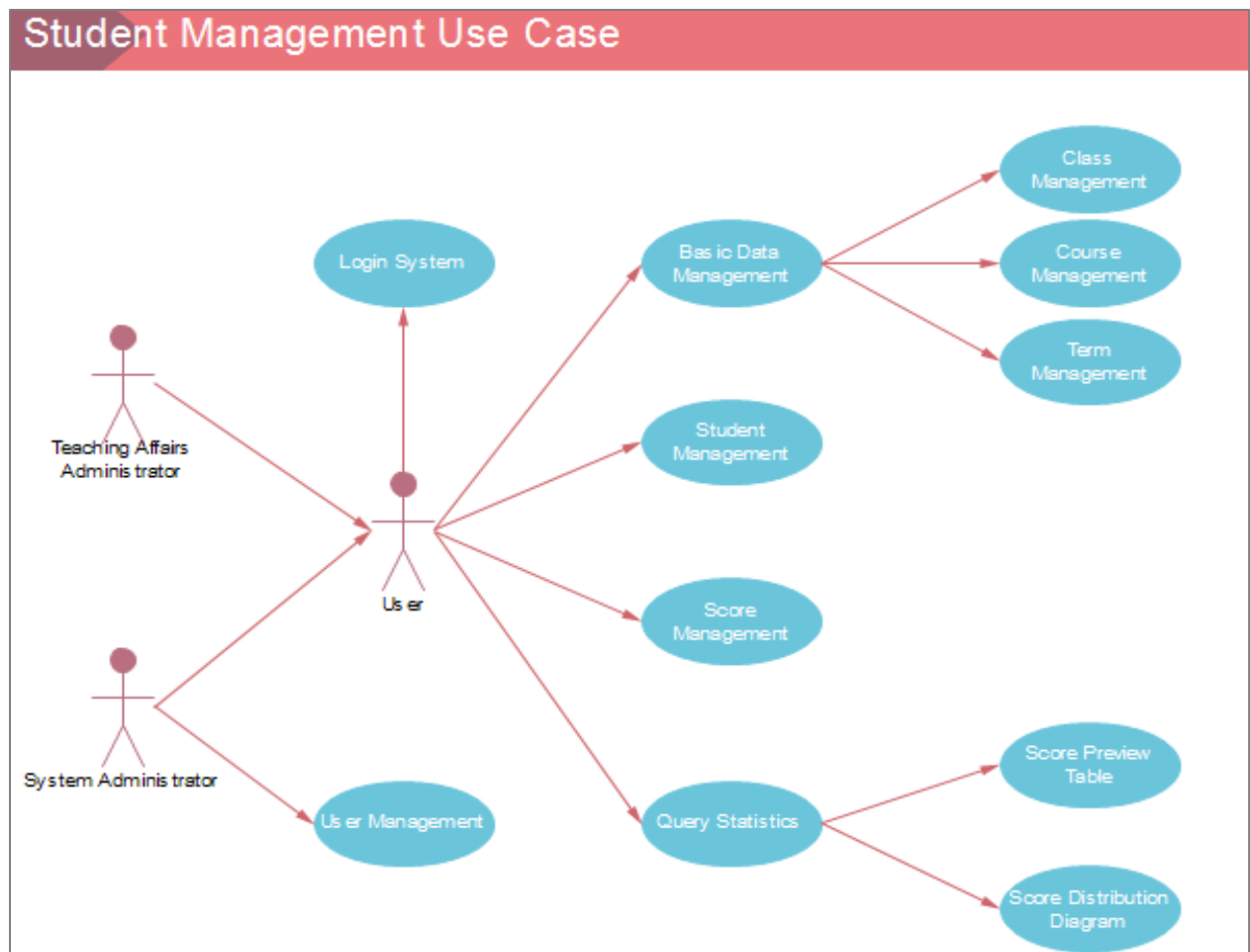


Fig. 4.6.3: Use Case Diagram for Student Management System

Use-case diagrams are helpful in the following situations:

- While gathering requirements, you can create use-case diagrams to capture the system requirements and to present to others what the system should do.
- During the analysis and design phases, you can use the use cases and actors from your use-case diagrams to identify the classes that the system requires.
- During the testing phase, you can use use-case diagrams to identify tests for the system.

The following topics describe model elements in use-case diagrams:

- Use Cases

A use case describes a function that a system performs to achieve the user's goal. A use case must yield an observable result that is of value to the user of the system.

- Actors

An actor represents a role of a user that interacts with the system that you are modelling. The user can be a human user, an organization, a machine, or another external system.

- Subsystems

In UML models, subsystems are a type of stereotyped component that represent independent, behavioural units in a system. Subsystems are used in class, component, and use-case diagrams to represent large-scale components in the system that you are modelling.

- **Data Flow Diagram:**

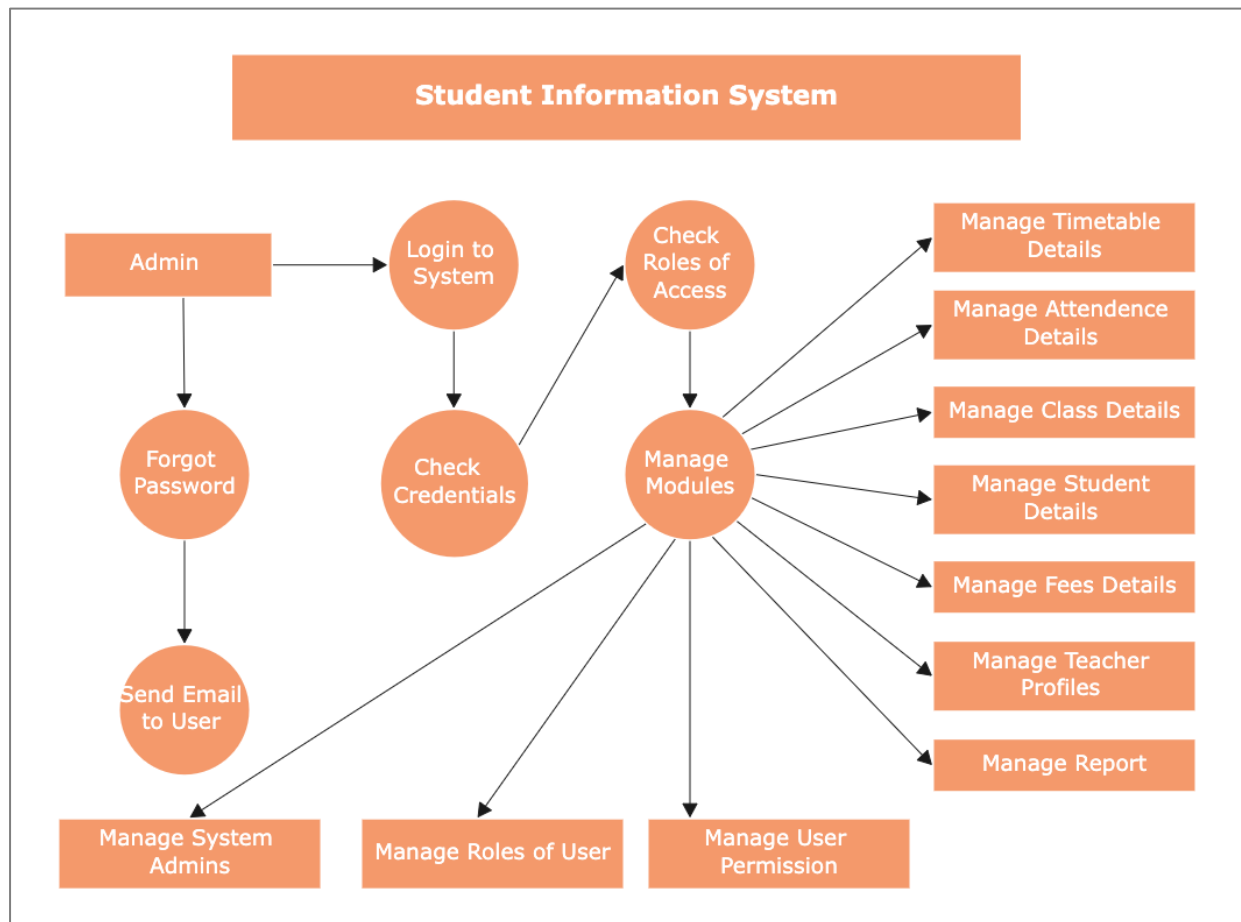


Fig. 4.6.4: Data Flow Diagram for Student Management System

- **Sequence Diagram:**

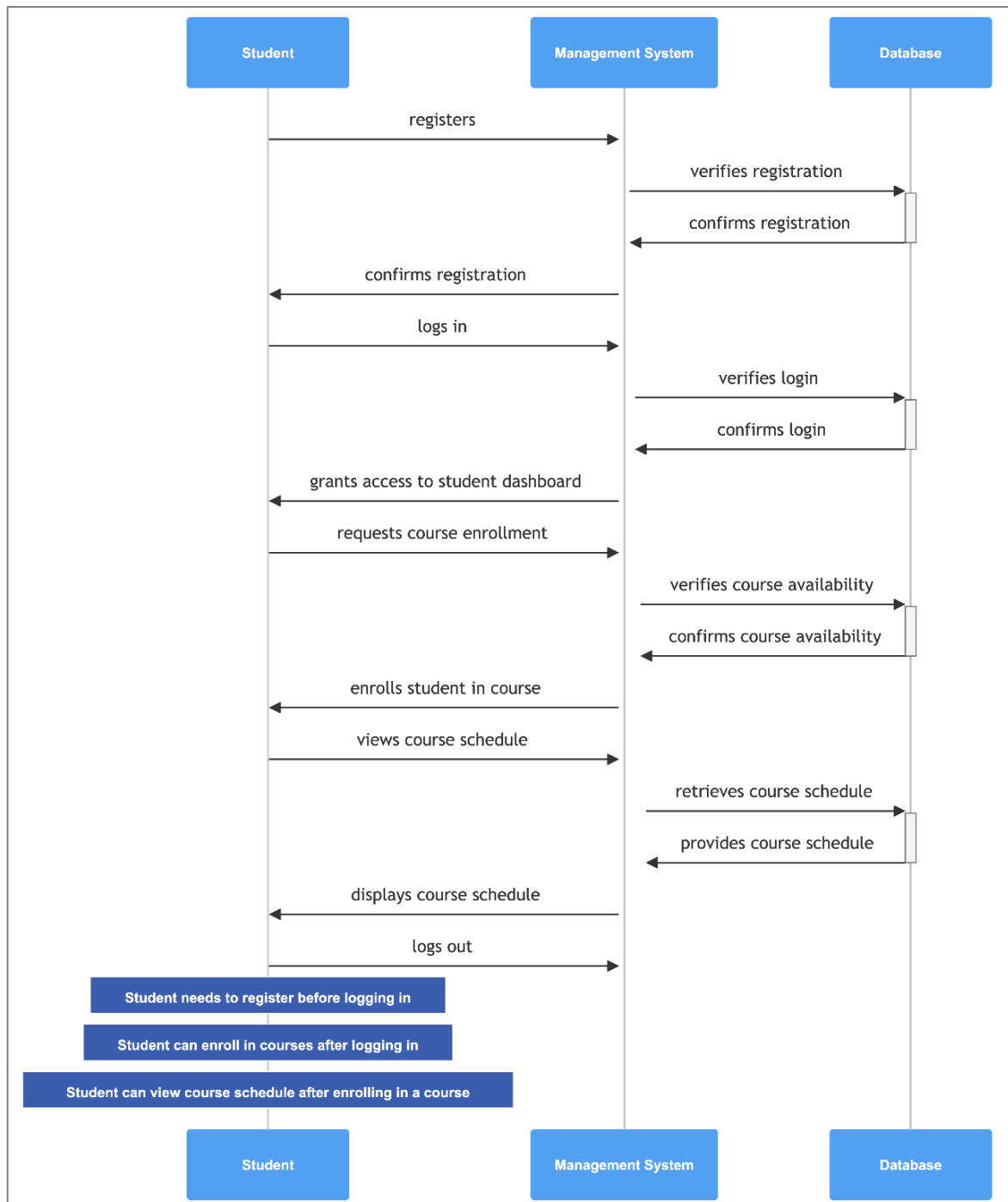


Fig. 4.6.5: Sequence Diagram for Student Management System

Chapter 5: Implementation and Testing

7.1 Implementation Approach

Implementation testing, also known as deployment testing, is an essential step in the software development life cycle. It involves verifying that the software system is installed and configured correctly, and it is working as intended in the production environment. The main objective of implementation testing is to ensure that the system is stable, reliable, and meets its requirements.

In a Student Management System (SMS), the implementation approach focuses on effectively managing student data, such as personal information, academic performance, attendance, and other administrative functions.

For the Student Management System, implementation testing can be done in several phases. The first step would be to verify that the system has been installed correctly on the production environment and all necessary components, such as the web server and database are properly configured. The next step would be verified that the system is working as intended by testing its functionality such as logging in, marking attendance, generating notes and integrating external systems.

Overall, implementation testing is critical step in the software development process that ensures the system is stable and reliable. It is important to thoroughly test the system before deploying it to production to avoid any issue or error that could impact the systems functionality or security.

5.2 Testing Approach

A test approach is test methodology execution of task, it characterizes how testing would be completed. Test approach has two systems:

- Proactive – It is a methodology where the structure procedure is started as right on time as conceivable so as to discover and fix the imperfections before the construct is made.
- Reactive – It is a methodology where the testing isn't begun until the structure and coding is finished completely.

Testing and validation are the most important steps after the implementation of the developed system. The software must be executed several times in order to find out the errors in the different modules of the system.

The system was designed according to the requirement of the system. But we are not 100% confident. The lack of confidence stems from several things. First the system deals with large number of states, complex logic and activities. So, some error might occur in the system.

Error may be software, which is known as “SOFTWARE ERROR” i.e. the software doesn't do what the requirement says. So, an exhaustive and thorough testing must be conducted to ascertain whether the system produces right results. The project guide and the user both did testing.

i. Unit Testing

The software units in a system are modules and routines that are assembled and integrated to perform a specific function. Unit testing focuses first on modules, independently of one another, to locate errors. This enables, to detect of errors in coding and logic that are contained within each module.

This testing includes entering data and ascertaining if the value matches to the type and size supported by MySQL. The various controls are tested to ensure that each performs its action as required.

ii. Integration Testing

Data can be lost across any interface, one module can hurt another, sub-functions when combined, may not produce the desired major functions. Integration testing is a systematic testing to discover errors associated within the interface.

The objective is to take unit-tested modules and build a program structure. All the modules are combined and tested as a whole. Here the Server module and Client module options are integrated and tested.

This testing provides the assurance that the application is well integrated functional unit with smooth transition of data.

iii. User Acceptance

User Acceptance is a crucial phase in the software testing lifecycle where the end users or clients test the system to verify if it meets their business requirements and is ready for deployment.

User Acceptance Testing typically occurs after the system has passed other types of testing, such as unit testing, integration testing, and system testing.

5.3. Validation Check

The basic objective of the requirement validation activity is to ensure that “Student Management System” reflects the actual requirements accurately and clearly.

Checking data for appropriateness and accuracy is called validating.

Data entry checks can be built into the database packages to prevent invalid data from being entered. For instance, in database, format checking (determining if the data are in the correct format) is done automatically because at the time of creating a database file.

Range checking (determining if the data fall within an acceptable range) and accuracy checking (making sure that an entry is possible) can be built by writing a simple program in database.

5.4 Test Cases

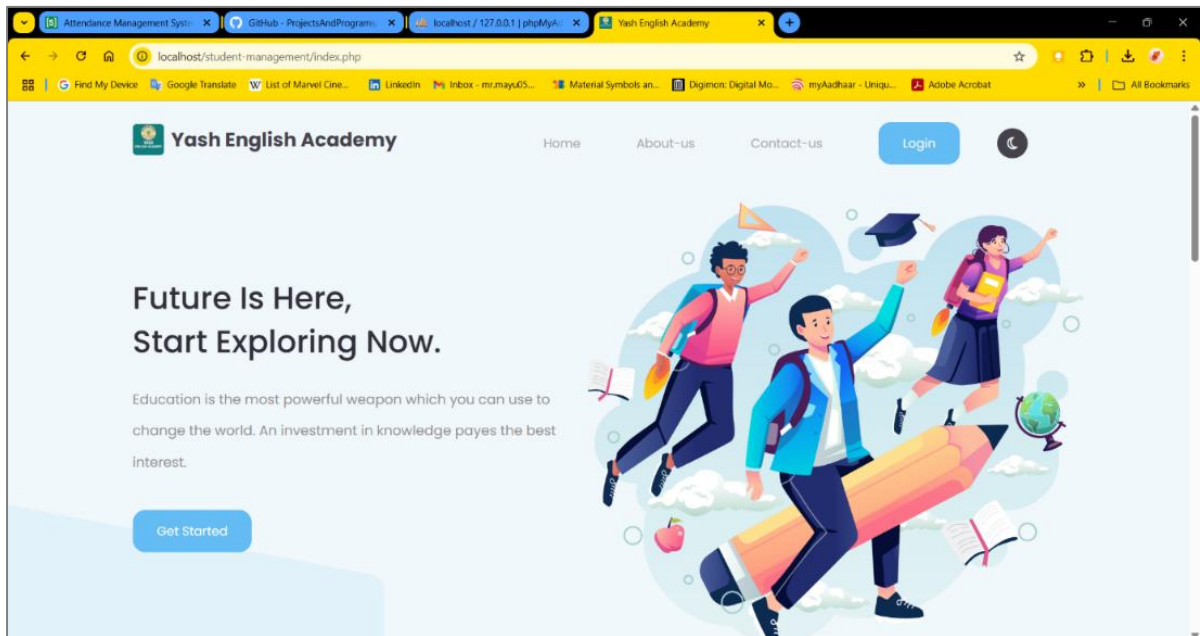
Test Case ID	Test Case Details	Test Steps	Expected Result	Actual Result	Status
TC-01	Test Student Login	1. Navigate to the login page. 2. Enter valid student credentials. 3. Click "Login".	Student should be logged in and redirected to their dashboard.	Student was successfully logged in and redirected to the dashboard.	Pass
TC-02	Test Login with Invalid Credentials	1. Navigate to the login page. 2. Enter invalid credentials. 3. Click "Login".	An error message should appear.	An error message will appear after attempting login with incorrect credentials.	Pass
TC-03	Test Adding Syllabus	1. Log in as Admin. 2. Navigate to the syllabus section. 3. Fill details. 4. Submit.	The syllabus should be added successfully, and it should appear in the list.	The syllabus successfully, and it is visible.	Pass
TC-04	Test Update Marks by Teacher	1. Log in as a teacher. 2. Navigate to the marks section. 3. Select a student. 4. Enter the marks. 5. Submit.	The marks should be updated successfully, and the student should be able to view it.	The marks were successfully updated for the student and the student could view it.	Pass

TC-05	Test Mark Attendance by Teacher	1. Log in as a teacher. 2. Navigate to the attendance section. 3. Select a class. 4. Mark students as present/absent. 5. Submit.	The attendance should be updated successfully for students.	The attendance should be updated successfully for the selected students.	Pass
TC-06	Test View Attendance by Student	1. Log in as a student. 2. Navigate to the attendance section. 3. View the attendance history.	The attendance for each course should be displayed.	The attendance for "Math 101" was displayed as present, and "History 201" was marked as absent.	Pass
TC-07	Test Admin View All Students	1. Log in as Admin. 2. Navigate to the student management section. 3. View the list of all students.	All registered students should be displayed in a list.	The list of all students was displayed with correct details such as name, ID, and registration status.	Pass
TC-08	Test Admin Delete Student Record	1. Log in as Admin. 2. Navigate to the student section. 3. Select a student. 4. Click "Delete".	The student record should be deleted from the system.	The student record was successfully deleted.	Pass

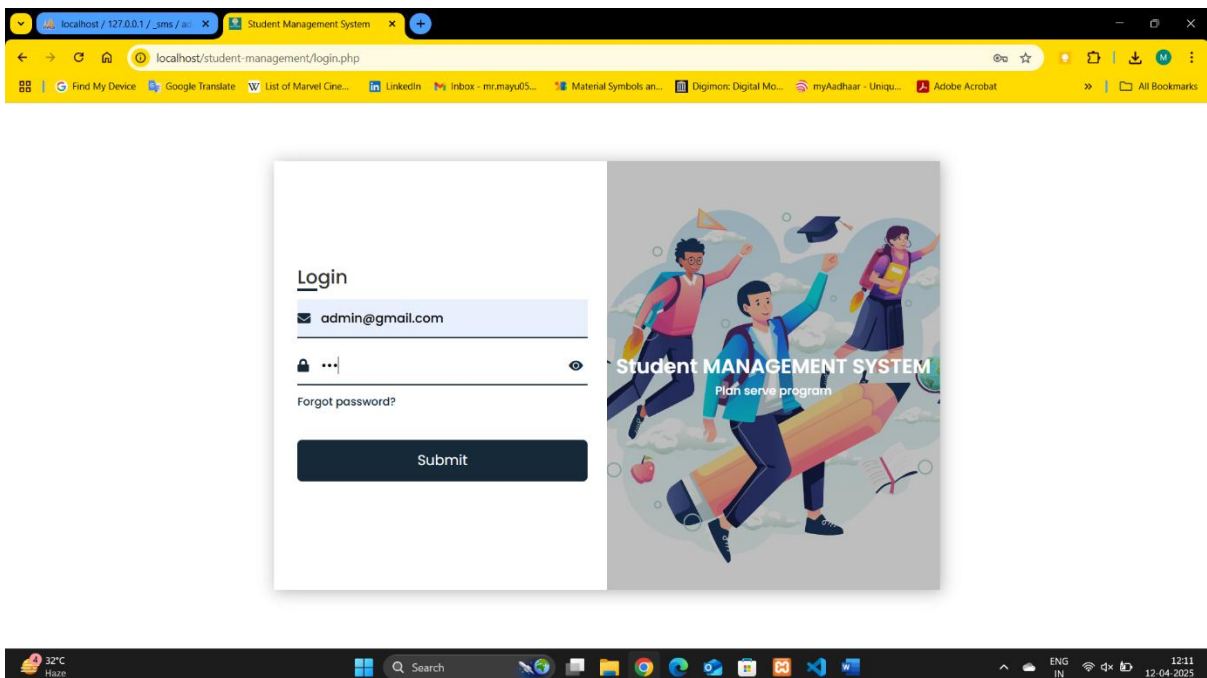
TC-09	Test Search for Student	1. Log in as Admin. 2. Navigate to the student section. 3. Enter a student name in the search bar. 4. Click "Search".	The system should display the student record that matches the search criteria.	The student "John Doe" appeared in the search results when searched by name.	Pass
TC-010	Test Logout Functionality	1. Log in as a student/teacher/admin. 2. Click on "Logout" button.	The user should be logged out and redirected to the login page.	The user was successfully logged out and redirected to the login page.	Pass

Chapter 6: Screen Layouts

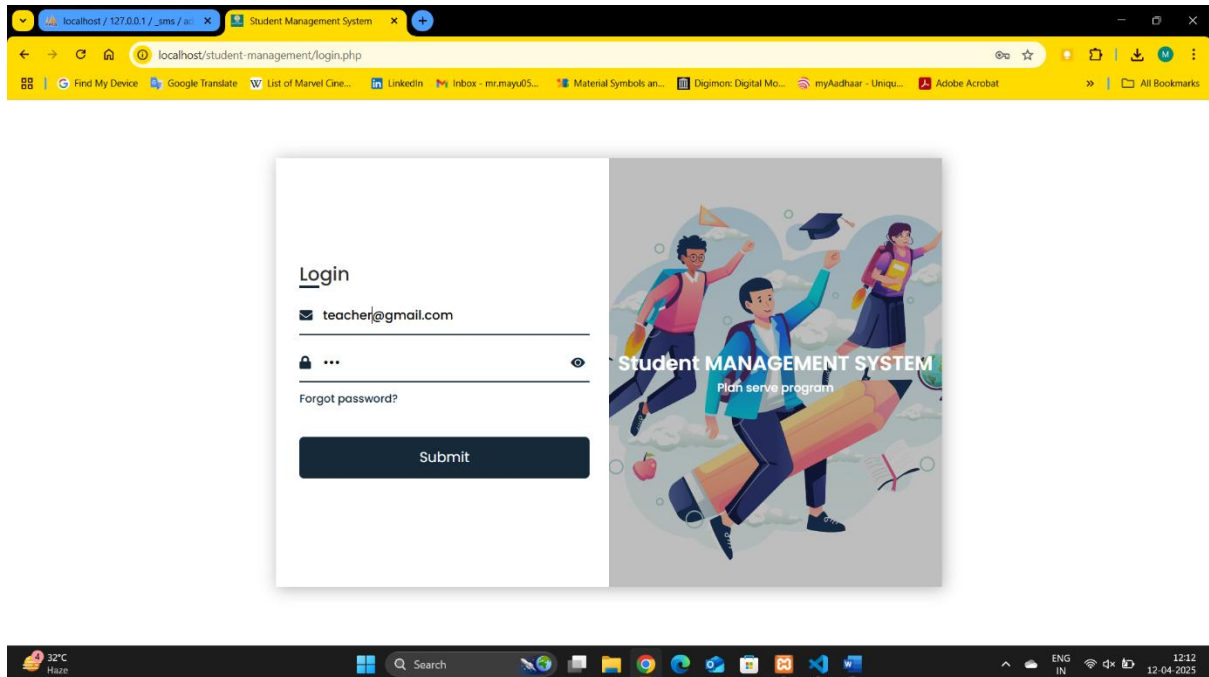
- **Home Page:**



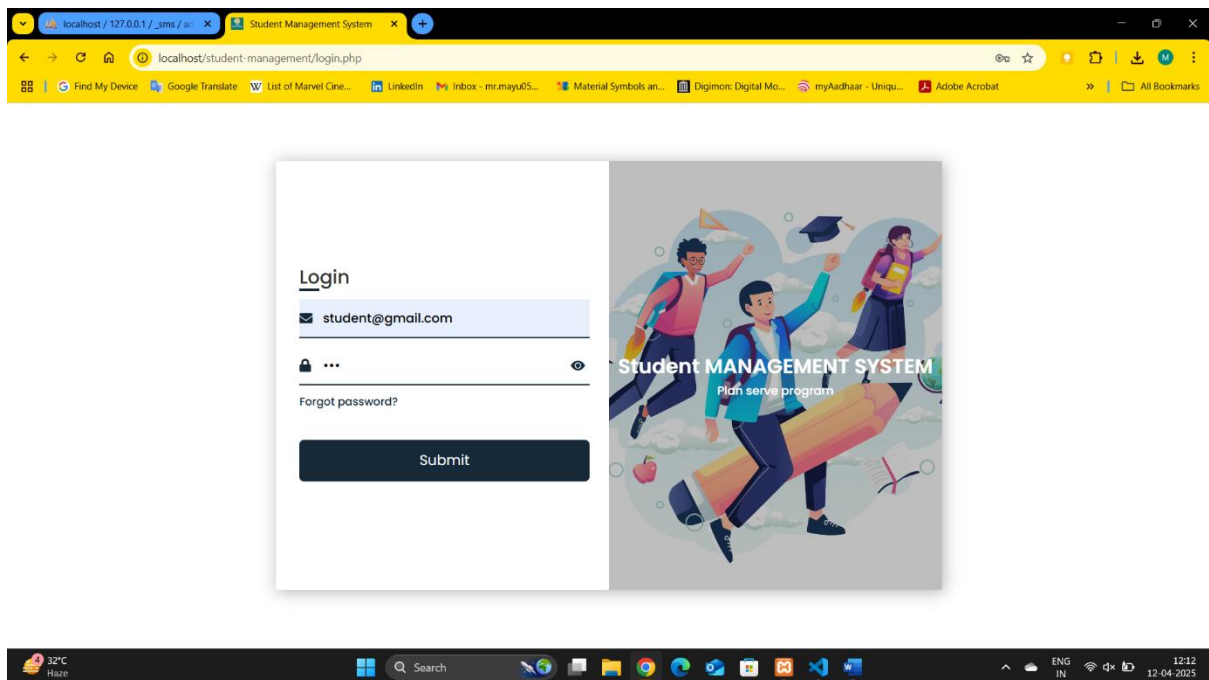
- **Admin Login Page:**



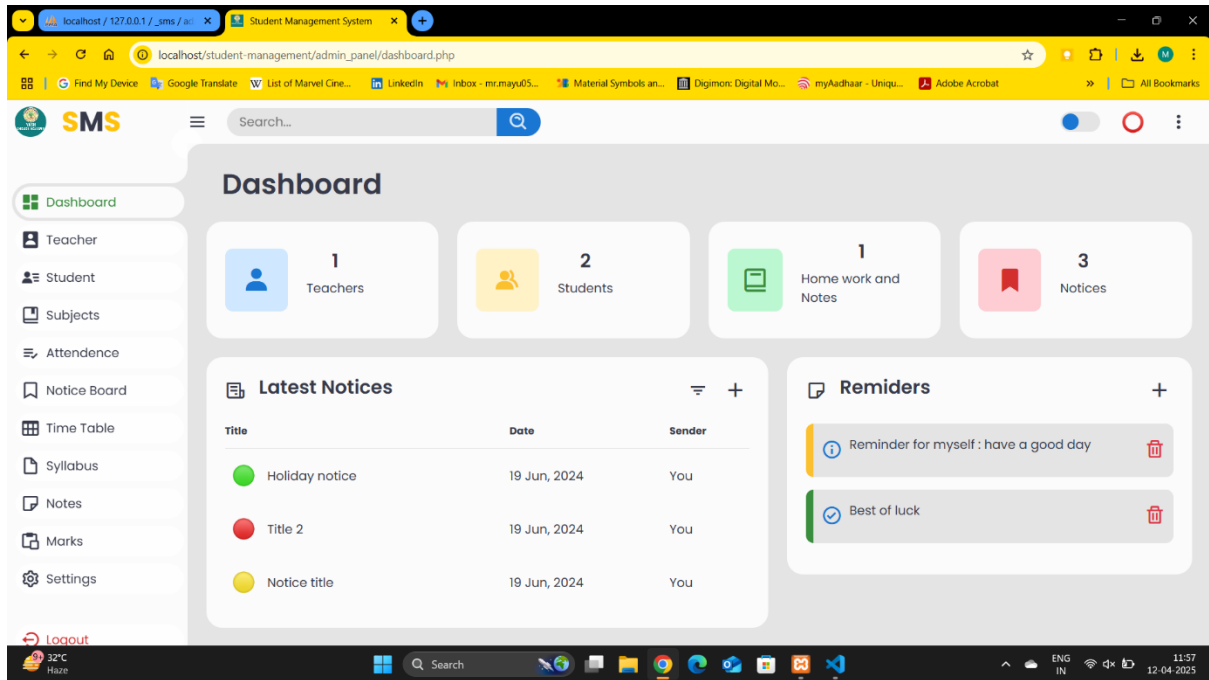
- **Teacher Login Page:**



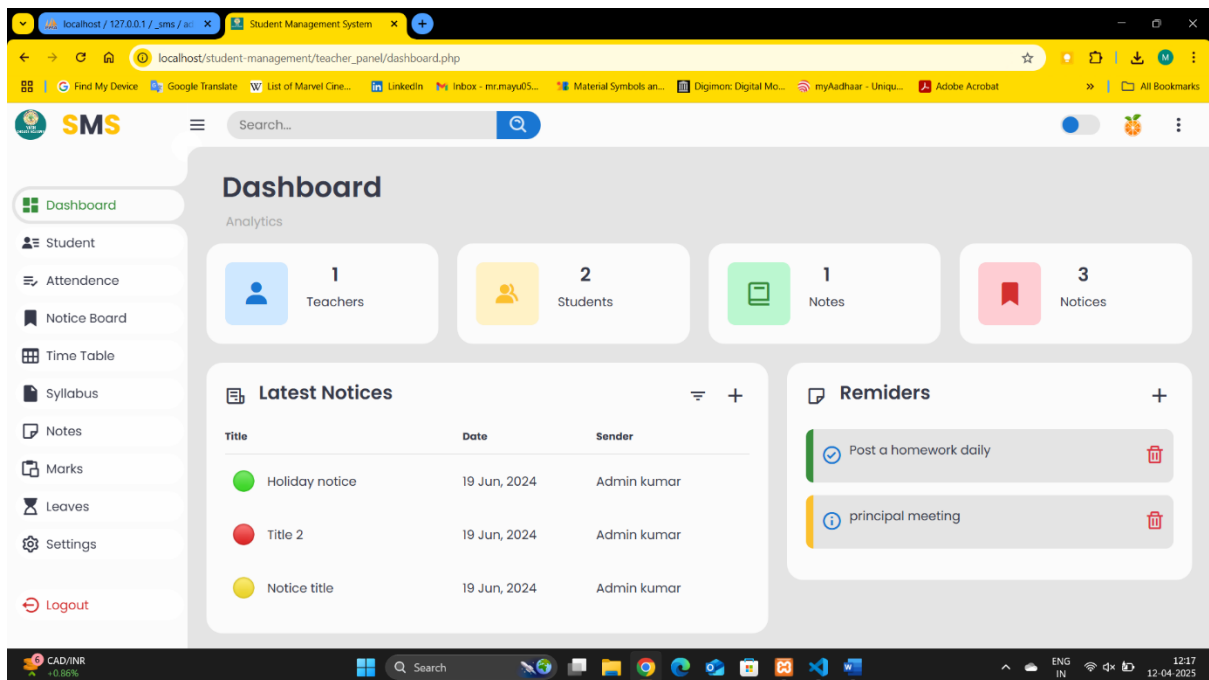
- **Student Login Page:**



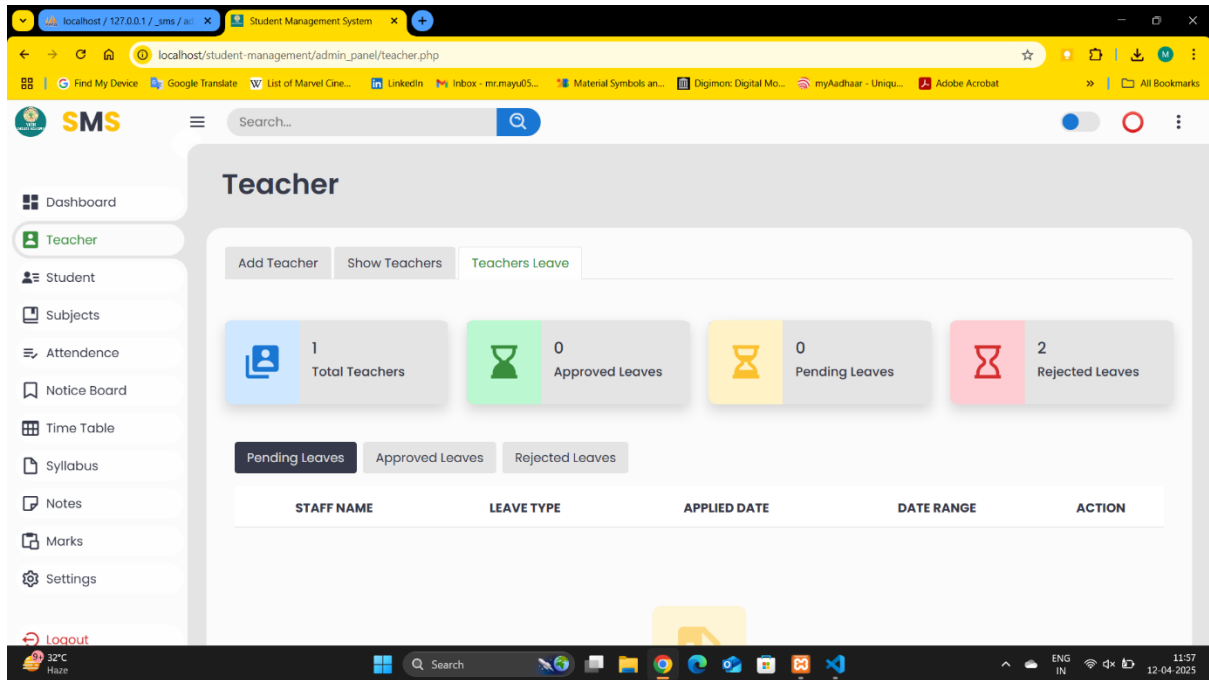
- **Admin Dashboard Page:**



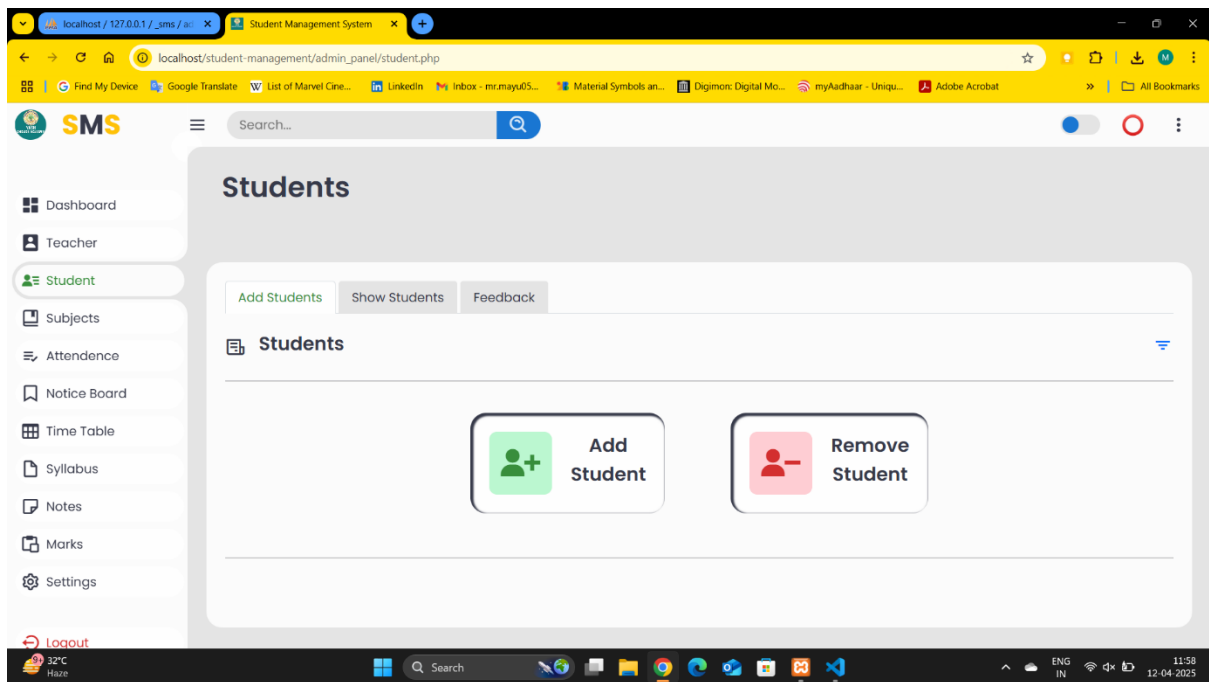
- **Teacher Dashboard Page:**



- **Teacher Page:**



- **Student Page:**



- **Subject Page:**

Subjects

Class: 12 (Commerce)

Find

Class 12c Subjects

#	Subject	Action
1.	Hindi	Edit Delete
2.	Commerce	Edit Delete

- **Attendance Page:**

Attendance

Take Attendance Date Wise Attendance

Information

Class: 12 (Commerce)

Section: A

Date: 12-04-2025

Find

Attendance Sheet

#	Roll No.	Name	Attendance
1.	S1718791292	Student kumar	Present

- **Time Table Page:**

The screenshot shows the 'Time Table' page for 'Class 12c A'. The page has a sidebar with navigation options: Dashboard, Teacher, Student, Subjects, Attendance, Notice Board, Time Table (selected), Syllabus, Notes, Marks, Settings, and Logout. The main content area displays a table for Monday's schedule. The table has columns for 'Period start', 'Period end', and 'Subject'. The schedule includes periods from 07:00 to 11:00, a lunch break (LUNCH) from 11:00 to 12:00, and periods from 12:00 to 03:00. The subjects listed are Hindi, English, Math, and Commerce.

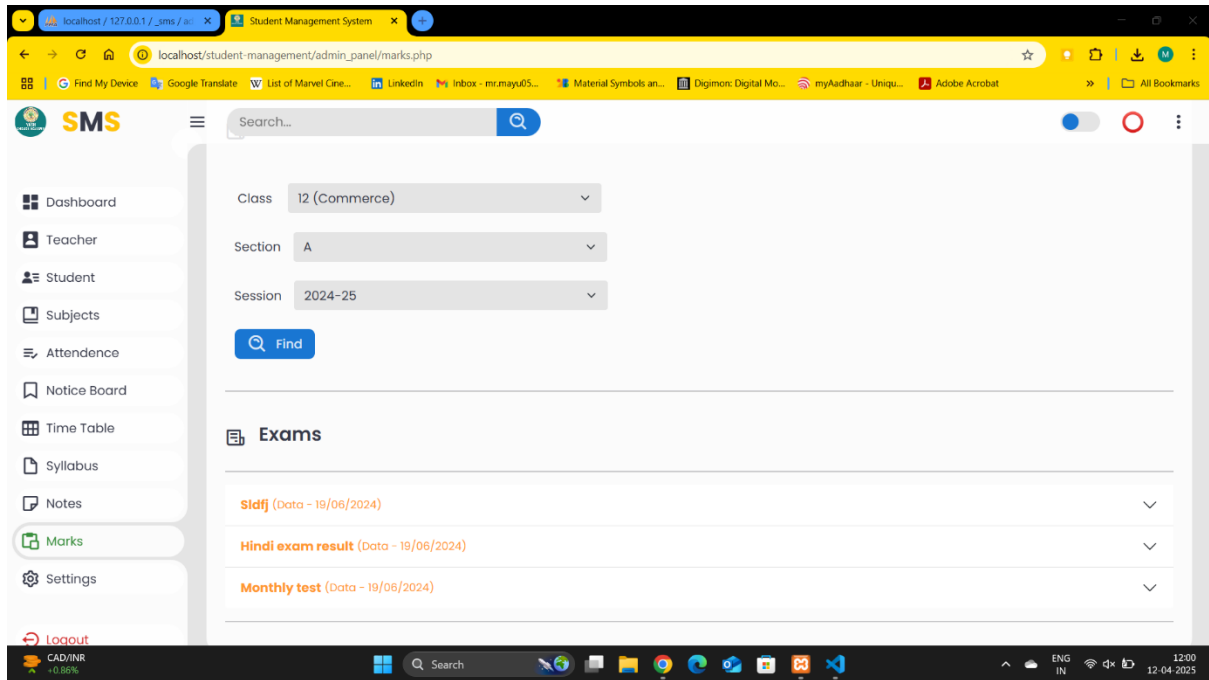
Period start	Period end	Subject
07:00	08:00	Hindi
08:00	09:00	English
09:00	10:00	Math
10:00	11:00	Hindi
11:00	12:00	English
LUNCH		
12:00	01:00	Commerce
01:00	02:00	Commerce
02:00	03:00	Hindi

- **Syllabus Page:**

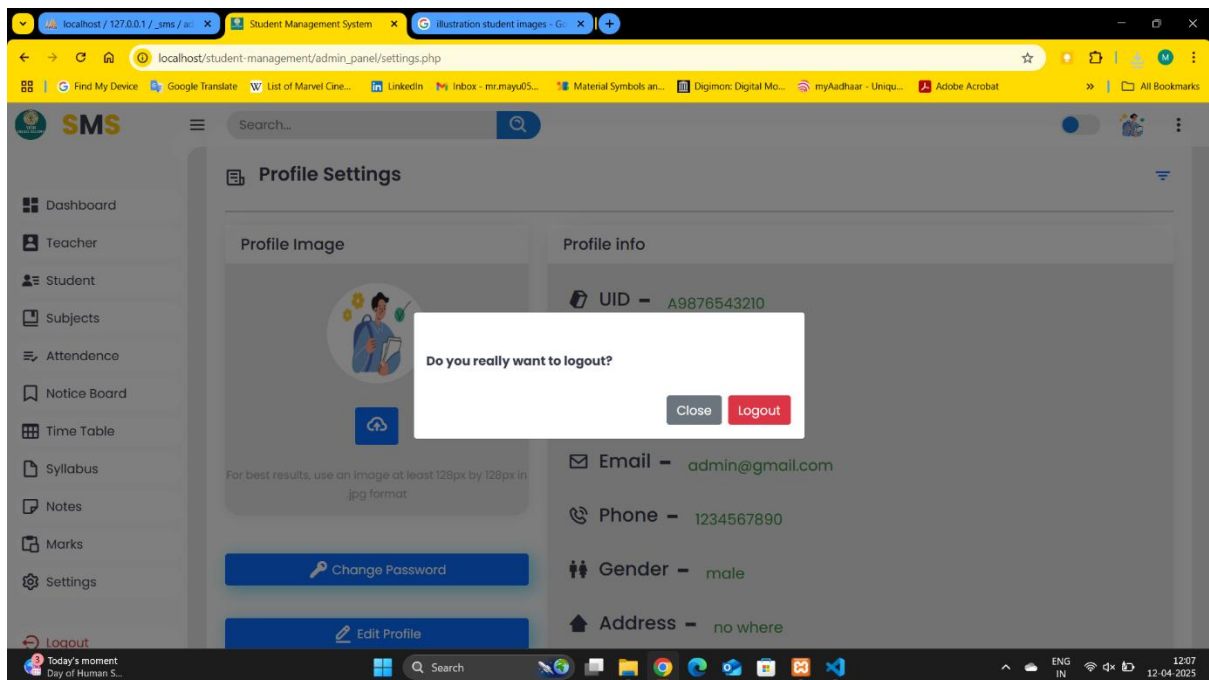
The screenshot shows the 'Syllabus' page in the SMS system. The sidebar is the same as the previous page. The main content area has a title 'Syllabus' and an 'Upload Syllabus' button. Below this, there is a dropdown menu for 'Class' set to '12 (Commerce)' and a 'Find' button. A table lists the subjects and their syllabi. The table has columns for '#', 'Subject', and 'Syllabus'. The subjects listed are Hindi and English. Each subject has a 'Download' button and an 'Upload' button.

#	Subject	Syllabus
1.	Hindi	Download Upload
2.	English	Download Upload

- **Marks Page:**



- **Logout Page:**



Chapter 7: Source Code

1. Home Page:

```
<div class="container mt-5">
  <div class="row">
    <div class="col-12 col-md-6 d-flex justify-content-center get-started" style="height:
550px;">
      <div class=" d-flex justify-content-center align-items-center">
        <div>
          <div class="big-title">
            <h1>Future is here,</h1>
            <h1>Start Exploring now.</h1>
          </div>
          <p class="text">
            Education is the most powerful weapon which you can use to change the world.
            An investment in knowledge payes the best interest.
          </p>
          <div class="cta">
            <a href="login.php" class="btn">Get started</a>
          </div>
        </div>
      </div>
    </div>
  </div>
```

2. About-us Page:

```
<div class="container mt-5">
  <div class="card border-0 shadow about-card">
    <div class="card-body mt-5 mb-2 mx-3 ">
      <h3 class="card-title">
```

```
<span class="text-primary icon-hover"><i class="fa-solid fa-paperclip"></i></span>
```

```
About us</h3>
```

```
<p class="card-text fs-5 text-justify">
```

Embark on your journey with courage and curiosity. Embrace the challenges, for they will become the milestones of your progress. Remember, every expert was once a beginner. Keep learning, stay persistent, and let your passion guide you. Your potential is limitless.

```
</div>
```

```
<div class="mb-4" style="width: 100%; display: flex; align-items: center; justify-content: center;">
```

```

```

```
</div>
```

```
</div>
```

```
</div>
```

3. Login Page:

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Yash English Academy</title>
```

```
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap.min.css">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css">
```

```
<link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.3.0/font/bootstrap-icons.css" />
```

```
<link rel="stylesheet" href="login-form-style.css">
```

```
<link rel="icon" type="image/x-icon" href="images/1.jfif">
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<input type="checkbox" id="flip">
```

```
<div class="cover">
```

```

<div class="front">
  
  <div class="text">
    <span class="text-1">SCHOOL MANAGEMENT SYSTEM<br></span>
    <span class="text-2">Plan serve program</span>
  </div>
</div>
<div class="forms">
  <div class="form-content">
    <div class="login-form">

      <div class="title" id='board-title'>Login</div>
      <div class="alert-box">
        <div class="alert alert-danger text-center mt-3" role="alert" id="error-msg">
          </div>
        </div>
        <form action="index.php" id="login-form" method="post">
          <div class="input-boxes">
            <div class="input-box">
              <i class="fas fa-envelope"></i>
              <input type="email" name="email" placeholder="Enter your email"
id='loginEmail' required>
            </div>
            <div class="input-box">
              <i class="fas fa-lock"></i>
              <input type="password" name="password" placeholder="Enter your password"
id="password" required>
              <i class="bi bi-eye-fill" style="margin-left:auto;margin-right: 6px;"
id="togglePassword"></i>
            </div>
            <div class="text"><a id="forgotpassword">Forgot password?</a></div>
            <div class="button input-box">

```

```

        <button type="submit" class="btn">
            Submit
        </button>
    </div>
</div>
</form>

```

4. Admin Page:

```

<div class="modal fade" id="edit-confirmation-modal" tabindex="-1" aria-
labelledby="exampleModalLabel"
    aria-hidden="true">
    <div class="modal-dialog modal-dialog-centered">
        <div class="modal-content">
            <div class="modal-header">
            </div>
            <div class="modal-body">
                <strong>Do you really want to Edit Student Details?</strong>
            </div>
            <div class="modal-footer">
                <button type="button" class="btn btn-secondary" data-bs-
dismiss="modal">Close</button>
                <button type="button" class="btn btn-danger" id="confirm-edit-
btn">Edit</button>
            </div>
        </div>
    </div>
</div>

<div class="modal fade" id="delete-confirmation-modal" tabindex="-1" aria-
labelledby="exampleModalLabel"
    aria-hidden="true">
    <div class="modal-dialog modal-dialog-centered">
        <div class="modal-content">
            <div class="modal-header">

```

```

</div>

<div class="modal-body">
    <strong>Do you really want to delete Student?</strong>
</div>

<div class="modal-footer">
    <button type="button" class="btn btn-secondary" data-bs-
dismiss="modal">Close</button>

    <button type="button" class="btn btn-danger"
onclick="deleteTeacherWithIdSeted()">Delete</button>
</div>
</div>
</div>
</div>

```

5. Teacher Page:

```
?php include('partials/_header.php') ?>
```

```
<!-- Sidebar -->
```

```
<?php include('partials/_sidebar.php') ?>
```

```
<input type="hidden" value="1" id="checkFileName">
```

```
<!-- End of Sidebar -->
```

```
<div class="modal fade" id="reminder-modal" tabindex="-1" aria-
labelledby="exampleModalLabel" aria-hidden="true">
```

```
<div class="modal-dialog modal-dialog-centered">
```

```
<div class="modal-content">
```

```
<div class="modal-header">
```

```
<h1 class="modal-title fs-5" id="exampleModalLabel">Add Reminder</h1>
```

```
<button type="button" class="close mr-2" data-bs-dismiss="modal" aria-
label="Close"><i
```

```
class='bx bx-x'></i></button>
```

```
</div>
```

```
<div class="modal-body">
```

```

        <div class="container mr-3 ml-3">
            <div class="alert alert-warning reminder-error" role="alert" style="min-
height: 50px;display: none;">
                Message can't be empty!
            </div>
            <div class="mb-3">
                <!-- <label for="exampleFormControlTextarea1" class="form-
label">Example textarea</label> -->
                <textarea class="form-control" id="reminder-msg"
rows="3"></textarea>
            </div>
        </div>

        <div class="modal-footer">
            <button type="button" class="btn btn-primary text-center _flex-container"
onclick="addReminder()"> <i
                class='bx bx-plus'></i>&nbsp;<strong>ADD</strong></button>
        </div>
    </div>
</div>

```

6. Dashboard:

```

<div class="modal fade" id="reminder-modal" tabindex="-1" aria-
labelledby="exampleModalLabel" aria-hidden="true">
    <div class="modal-dialog modal-dialog-centered">
        <div class="modal-content">
            <div class="modal-header">
                <h1 class="modal-title fs-5" id="exampleModalLabel">Add Reminder</h1>
                <button type="button" class="close mr-2" data-bs-dismiss="modal" aria-
label="Close"><i

```

```

        class='bx bx-x'></i></button>
</div>
<div class="modal-body">
    <div class="container mr-3 ml-3">
        <div class="alert alert-warning reminder-error" role="alert" style="min-height:
50px;display: none;">
            Message can't be empty!
        </div>
        <div class="mb-3">
            <!-- <label for="exampleFormControlTextarea1" class="form-label">Example
textarea</label> -->
            <textarea class="form-control" id="reminder-msg" rows="3"></textarea>
        </div>
    </div>
    <div class="modal-footer">
        <button type="button" class="btn btn-primary text-center _flex-container"
onclick="addReminder()"> <i
            class='bx bx-plus'></i>&nbsp;<strong>ADD</strong></button>
        </div>
    </div>
</div>

```

Chapter 8: Conclusions

8.1 Conclusion

It's usually a good idea to go with a student information system that's built on current system architecture to keep up with changing needs. This system should include well organized data coding and clearly defined business applications.

The system's overview elucidates the convenience of exact data delivery at the tip of your fingertips, increasing student retention and teaching them how to manage their time effectively.

The proposed method is efficient and user-friendly, based on the results of the experiments and tests.

In comparison to current methods of managing academic institutions, this project, which produces centralised software, makes work administration and management easier and gives full information about the issue of users' interest with just one mouse click.

Easy-to-use user interface centralised software can be offered to the educational institution, allowing all services linked with the university to interact with one another and share data.

The user will be able to access the resources from afar because this is a ReST API. Because the application is built with a micro service architecture and agile methodology, services can be added in the future.

8.2 Future Work

The project's future potential is enormous. In the future, the project could be deployed on an intranet. Because it is quite versatile in terms of expansion, the project can be upgraded in the near future as and when the need arises.

The customer may now manage and thus run the complete task in a lot better, accurate, and error-free manner now that the planned database Space Manager software is ready and fully functional.

In the future students may also be able to post or download notes in the future. There will be a few little modifications here and there to make the app more visually appealing, as well as statistics, tracking, and analytics. For significantly more, the entire project will be made available as an Android app.

Artificial intelligence's contribution to several fields is growing all the time. So, in the future we will be using some machine learning algorithms in application, to make it more efficient.

Bar code reader-based attendance system will be used in future. Since, it is more secure and efficient for teacher to access the details of any student.

In future student can be login using the fingerprint or photo. We should continue to think on what is happening and consider whether things could be made better.

Chapter 9: References

1. www.youtube.com
2. www.google.com
3. <https://www.scribd.com>
4. <https://studentprojectguide.com>
5. <https://www.studocu.com>
6. <https://www.academia.edu>